

Light

MCQ

1. Light is a type of

- a) Sound wave
- b) Mechanical wave
- c) Electromagnetic radiation
- d) Heat energy

Answer: c) Electromagnetic radiation

2. Light helps human beings to

- a) Hear sounds
- b) Feel heat
- c) See objects
- d) Measure time

Answer: c) See objects

3. Light is defined as a form of energy that

- a) Produces heat only
- b) Enables us to see surroundings
- c) Produces sound
- d) Creates force

Answer: b) Enables us to see surroundings

4. Objects which emit light are called

- a) Reflectors
- b) Mirrors
- c) Light sources
- d) Lenses

Answer: c) Light sources

5. The two main sources of light are

- a) Bright and dim
- b) Hot and cold
- c) Natural and artificial
- d) Solid and liquid

Answer: c) Natural and artificial

6. The primary natural source of light in our solar system is

- a) Moon
- b) Stars
- c) Sun
- d) Fire

Answer: c) Sun

7. Light produced by living organisms is called

- a) Fluorescence
- b) Incandescence
- c) Bioluminescence
- d) Phosphorescence

Answer: c) Bioluminescence

8. Fireflies produce light due to

- a) Heat
- b) Electricity
- c) Chemical reactions
- d) Reflection

Answer: c) Chemical reactions

9. Artificial sources of light are

- a) Natural

b) Man-made

c) Living

d) Celestial

Answer: b) Man-made

10. Which of the following is an artificial source of light?

a) Sun

b) Firefly

c) Candle

d) Star

Answer: c) Candle

11. Incandescent sources emit light due to

a) Chemical reaction

b) Electric discharge

c) High temperature

d) Reflection

Answer: c) High temperature

12. Candle flame is an example of

a) Gas discharge source

b) Luminescent source

c) Incandescent source

d) Natural source

Answer: c) Incandescent source

13. Luminescence is the emission of light

a) With high heat

b) Without appreciable heat

c) Only in gases

d) Only in solids

Answer: b) Without appreciable heat

14. Fluorescent tube light is a

a) Natural source

b) Incandescent source

c) Luminescent source

d) Reflection source

Answer: c) Luminescent source

15. Neon lamp is an example of

a) Incandescent source

b) Gas discharge source

c) Natural source

d) Luminescent solid

Answer: b) Gas discharge source

16. Light travels in a

a) Curved path

b) Zig-zag path

c) Straight line

d) Circular path

Answer: c) Straight line

17. Straight-line travel of light is called

a) Reflection

b) Refraction

c) Rectilinear propagation

d) Dispersion

Answer: c) Rectilinear propagation

18. Device used to demonstrate rectilinear propagation of light is

- a) Microscope
- b) Telescope
- c) Pinhole camera
- d) Periscope

Answer: c) Pinhole camera

19. A pinhole camera consists of

- a) Lens and mirror
- b) Light-tight box and pinhole
- c) Prism and screen
- d) Bulb and battery

Answer: b) Light-tight box and pinhole

20. Image formed in a pinhole camera is

- a) Real and inverted
- b) Virtual and erect
- c) Enlarged
- d) Laterally inverted

Answer: a) Real and inverted

21. Reflection is the

- a) Absorption of light
- b) Bending of light
- c) Bouncing back of light
- d) Splitting of light

Answer: c) Bouncing back of light

22. Reflection occurs from

- a) Rough surfaces only

b) Smooth polished surfaces

c) Transparent materials

d) Translucent materials

Answer: b) Smooth polished surfaces

23. A mirror is made of

a) Plastic

b) Metal

c) Glass with shiny surface

d) Rubber

Answer: c) Glass with shiny surface

24. The ray that falls on the mirror is called

a) Reflected ray

b) Normal ray

c) Incident ray

d) Emergent ray

Answer: c) Incident ray

25. The ray that bounces back from the mirror is

a) Incident ray

b) Reflected ray

c) Normal ray

d) Parallel ray

Answer: b) Reflected ray

26. The imaginary line perpendicular to the mirror is called

a) Tangent

b) Axis

c) Normal

d) Diameter

Answer: c) Normal

27. Angle between incident ray and normal is

a) Angle of reflection

b) Angle of deviation

c) Angle of incidence

d) Right angle

Answer: c) Angle of incidence

28. Angle of incidence is always equal to

a) Angle of deviation

b) Angle of reflection

c) Angle of refraction

d) Zero

Answer: b) Angle of reflection

29. According to laws of reflection, rays lie in

a) Different planes

b) Same plane

c) Parallel planes

d) Random planes

Answer: b) Same plane

30. Regular reflection occurs on

a) Rough surface

b) Uneven surface

c) Smooth surface

d) Dusty surface

Answer: c) Smooth surface

31. Reflection from a plane mirror is an example of

- a) Irregular reflection
- b) Diffused reflection
- c) Regular reflection
- d) Absorption

Answer: c) Regular reflection

32. Irregular reflection occurs on

- a) Smooth surface
- b) Rough surface
- c) Polished surface
- d) Mirror

Answer: b) Rough surface

33. A collection of light rays is called

- a) Wave
- b) Spectrum
- c) Beam
- d) Shadow

Answer: c) Beam

34. Laser beam is an example of

- a) Diverging beam
- b) Converging beam
- c) Parallel beam
- d) Curved beam

Answer: c) Parallel beam

35. Light spreading in all directions forms

- a) Parallel beam

b) Converging beam

c) Diverging beam

d) Focused beam

Answer: c) Diverging beam

36. Magnifying lens produces

a) Parallel beam

b) Diverging beam

c) Converging beam

d) Random beam

Answer: c) Converging beam

37. Speed of light in air or vacuum is

a) 3×10^6 m/s

b) 3×10^8 m/s

c) 3×10^{10} m/s

d) 300 m/s

Answer: b) 3×10^8 m/s

38. Speed of light is the

a) Slowest speed

b) Variable speed

c) Highest speed in nature

d) Average speed

Answer: c) Highest speed in nature

39. Light interacts with matter by

a) Absorption

b) Reflection

c) Refraction

d) All of the above

Answer: d) All of the above

40. Objects that do not allow light to pass are

a) Transparent

b) Translucent

c) Opaque

d) Reflective

Answer: c) Opaque

41. Wooden door is an example of

a) Transparent object

b) Translucent object

c) Opaque object

d) Luminous object

Answer: c) Opaque object

42. Objects that allow light to pass completely are

a) Opaque

b) Translucent

c) Transparent

d) Reflective

Answer: c) Transparent

43. Window glass is a

a) Opaque material

b) Translucent material

c) Transparent material

d) Luminous source

Answer: c) Transparent material

44. Butter paper allows light partially and is

- a) Transparent
- b) Opaque
- c) Translucent
- d) Reflective

Answer: c) Translucent

45. Shadow is formed when light falls on

- a) Transparent object
- b) Translucent object
- c) Opaque object
- d) Luminous object

Answer: c) Opaque object

46. Region of total darkness in a shadow is

- a) Penumbra
- b) Umbra
- c) Cone shadow
- d) Reflection

Answer: b) Umbra

47. Region of partial darkness is called

- a) Umbra
- b) Image
- c) Penumbra
- d) Focus

Answer: c) Penumbra

48. Solar eclipse occurs when

- a) Earth comes between Sun and Moon

- b) Moon comes between Earth and Sun
- c) Sun comes between Earth and Moon
- d) Stars block sunlight

Answer: b) Moon comes between Earth and Sun

49. Lunar eclipse occurs when

- a) Moon comes between Earth and Sun
- b) Earth comes between Sun and Moon
- c) Sun comes between Earth and Moon
- d) Clouds cover moon

Answer: b) Earth comes between Sun and Moon

50. It is safe to view

- a) Solar eclipse directly
- b) Lunar eclipse directly
- c) Both eclipses directly
- d) None

Answer: b) Lunar eclipse directly

51. Solar eclipse occurs on

- a) Full moon day
- b) New moon day
- c) Half moon day
- d) Any day

Answer: b) New moon day

52. Lunar eclipse occurs on

- a) New moon day
- b) Full moon night
- c) Half moon day

d) Any night

Answer: b) Full moon night

53. Image formed by a plane mirror is

a) Real

b) Inverted

c) Virtual and erect

d) Enlarged

Answer: c) Virtual and erect

54. Image distance behind mirror is

a) Greater than object distance

b) Less than object distance

c) Equal to object distance

d) Zero

Answer: c) Equal to object distance

55. Plane mirror produces image of

a) Same size as object

b) Smaller than object

c) Larger than object

d) No image

Answer: a) Same size as object

56. Lateral inversion means

a) Image upside down

b) Left and right interchanged

c) Image blurred

d) Image enlarged

Answer: b) Left and right interchanged

57. Image that can be obtained on a screen is

- a) Virtual image
- b) Erect image
- c) Real image
- d) Laterally inverted image

Answer: c) Real image

58. Image formed by plane mirror is

- a) Real
- b) Captured on screen
- c) Virtual
- d) Inverted

Answer: c) Virtual

59. Visible light wavelength range is

- a) 100–200 nm
- b) 200–300 nm
- c) 400–700 nm
- d) 700–900 nm

Answer: c) 400–700 nm

60. VIBGYOR represents

- a) Seven sounds
- b) Seven energies
- c) Seven colours
- d) Seven mirrors

Answer: c) Seven colours

61. Colour with shortest wavelength is

- a) Red

b) Yellow

c) Green

d) Violet

Answer: d) Violet

62. Colour with longest wavelength is

a) Violet

b) Indigo

c) Blue

d) Red

Answer: d) Red

63. White light is perceived when

a) Only red enters eye

b) Only blue enters eye

c) All colours enter eye together

d) No light enters eye

Answer: c) All colours enter eye together

64. Prism is made of

a) Opaque material

b) Reflective metal

c) Transparent material

d) Rubber

Answer: c) Transparent material

65. Splitting of white light into colours is called

a) Reflection

b) Refraction

c) Dispersion

d) Absorption

Answer: c) Dispersion

66. Newton disc proves that

a) Light travels fast

b) Light is straight

c) White light is made of seven colours

d) Reflection occurs

Answer: c) White light is made of seven colours

67. Newton disc appears white when

a) Kept still

b) Heated

c) Rotated fast

d) Covered

Answer: c) Rotated fast

68. Yellow gelatin paper allows only

a) Red light

b) Blue light

c) Yellow light

d) All colours

Answer: c) Yellow light

69. Mixing colours of light to form new colours is

a) Dispersion

b) Reflection

c) Synthesis of colour

d) Absorption

Answer: c) Synthesis of colour

70.Primary colours of light are

- a) Cyan, Magenta, Yellow
- b) Red, Green, Blue
- c) Violet, Indigo, Blue
- d) Yellow, Orange, Red

Answer: b) Red, Green, Blue

71.Additive colour mixing uses

- a) Pigments
- b) Paint
- c) Light
- d) Ink

Answer: c) Light

72.Equal mixing of RGB produces

- a) Black
- b) Yellow
- c) White
- d) Grey

Answer: c) White

73.Additive colour mixing is used in

- a) Printing
- b) Painting
- c) Television screens
- d) Cloth dyeing

Answer: c) Television screens

74.Subtractive colour mixing removes

- a) Heat

- b) Light
- c) Sound
- d) Energy

Answer: b) Light

75. Primary colours in subtractive mixing are

- a) RGB
- b) VIBGYOR
- c) CMY
- d) Black and white

Answer: c) CMY

76. Subtractive colour mixing is used in

- a) Cameras
- b) TV screens
- c) Printing
- d) Lasers

Answer: c) Printing

77. Plane mirror image is

- a) Real and inverted
- b) Virtual and upright
- c) Enlarged
- d) Reduced

Answer: b) Virtual and upright

78. Image in pinhole camera is

- a) Virtual
- b) Upright
- c) Real and inverted

d) Same size

Answer: c) Real and inverted

79. Light reflected from pond water shows

a) Absorption

b) Reflection

c) Refraction

d) Dispersion

Answer: b) Reflection

80. Light cannot bend its path by itself because of

a) Dispersion

b) Rectilinear propagation

c) Reflection

d) Absorption

Answer: b) Rectilinear propagation